



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
22-08-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S & I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

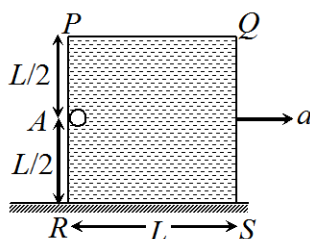
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A small solid ball of density ρ is held inside at point A a cubical container of side L , filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration a horizontally and the ball is released from point A simultaneously, then



(A) For ball to hit the top of container at end Q, $a = 3g$

(B) For ball to hit the top of container at end Q, $a = 2g$

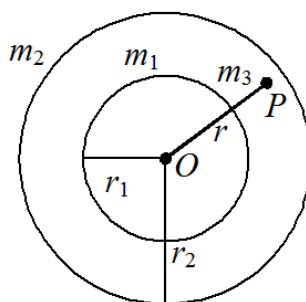
(C) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{L}{3g}}$

(D) Ball hits the top of container at end Q after a time $t = \sqrt{\frac{2L}{3g}}$

2. $Y(x, t) = \frac{0.8}{[(4x + 5t)^2 + 5]}$ represents a moving pulse where x and y are in metres and t in second.

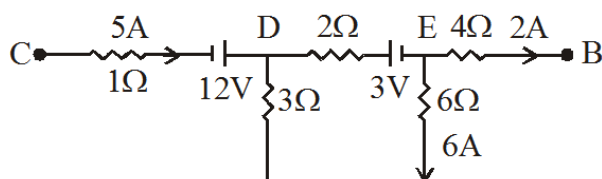
Then :

- (A) pulse is moving in positive x -direction.
 (B) in 2s it will travel a distance of 2.5 m.
 (C) maximum displacement of particles is 0.16 m.
 (D) it is a symmetric pulse at $t = 0$.
3. Two concentric spherical shells masses m_1 and m_2 and radii r_1 and r_2 . Then



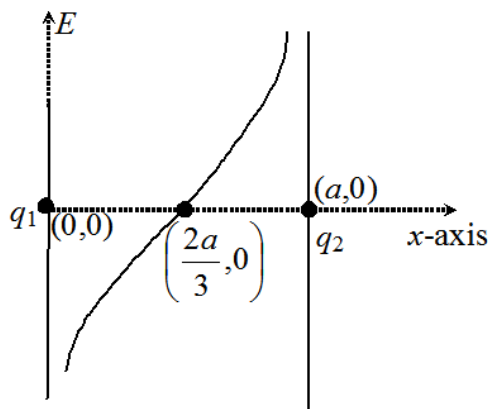
- (A) outer shell will have no contribution in gravitational field at point P.
 (B) force on P is directed towards O.
 (C) force on P is $\frac{Gm_1 m_2}{r^2}$.
 (D) force on P is $\frac{Gm_1 m_3}{r^2}$.

4. A part of the circuit is shown in the figure

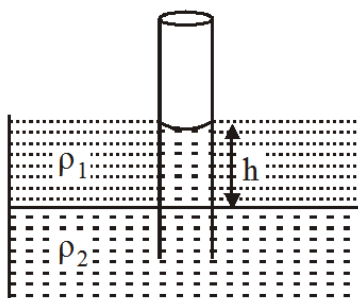


- (A) power dissipated in 3Ω resistance is 27 watt
 (B) $V_C - V_B = 20$ V
 (C) $V_C - V_D = 20$ V
 (D) power dissipated in 6Ω resistance is 36 watt

5. If the electric field E is plotted, with distance from q_1 along the line joining of two, then (E is positive along +ve x-axis) it looks as shown in figure. From the plot we can say that



- (A) q_1 and q_2 both are negative
(B) $\left| \frac{q_1}{q_2} \right| = 4$
(C) q_1 is positive and q_2 is negative
(D) $\left| \frac{q_1}{q_2} \right| = \frac{1}{4}$
6. A container is partially filled with a liquid of density ρ_2 . A capillary tube of radius ' r ' is vertically inserted in this liquid. Now another liquid of density ρ_1 ($\rho_1 < \rho_2$) is slowly poured in the container to a height h as shown. There is only denser liquid in the capillary tube. The rise of denser liquid in the capillary tube is also h . (Assume zero contact angle)



- (A) the surface tension of heavier liquid is $2\pi r(\rho_2 - \rho_1)gh$
(B) the surface tension of heavier liquid is $\frac{r}{2}(\rho_2 - \rho_1)gh$
(C) liquid level in capillary will be lower than free surface of lighter liquid if surface tension is zero.
(D) liquid level in capillary will be equal to the free surface of lighter liquid if surface tension is zero.

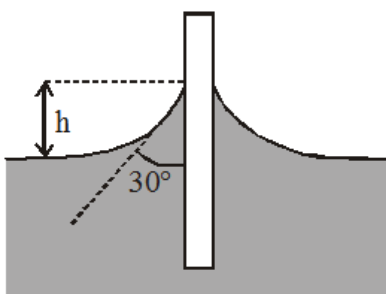
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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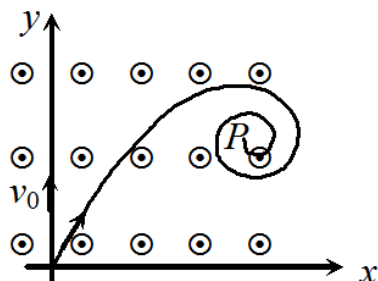
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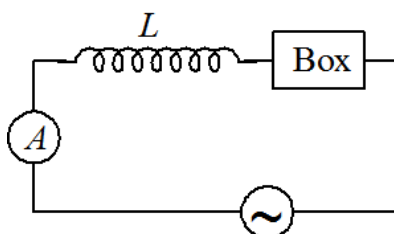
1. A rod of mass 'M' and length '2L' is suspended at its middle by a wire. It exhibits torsional oscillations; If two masses each of 'm' are attached at distance 'L/2' from its centre on both sides, it reduces the oscillation frequency by 20%. The value of ratio m/M is close to:
2. Two concentric coplanar circular loops made of wire, with resistance per unit length $10\Omega/\text{m}$ have diameters 0.2m and 2m. A time varying potential difference $(4 + 2.5t)$ volt is applied to the larger loop. The current in the smaller loop is $n \times 10^{-10}$ A, determine the value of n.
3. Voltage rating of a parallel plate capacitor is 500 V. Its dielectric can withstand a maximum electric field of 10^6 V/m. The plate area is 10^{-4}m^2 . What is the dielectric constant if the capacitance is 15pF ? (given $\epsilon_0 = 8.86 \times 10^{-12} \text{C}^2 \text{m}^{-2}$)
4. A thin sheet is inserted vertically inside a liquid of density ρ and surface tension T. The liquid rises by height 'h' close to sheet making angle of 30° with sheet, as shown in the figure. The value of 'h' in mm is _____. [$\rho = 10^3 \text{kg/m}^3$, $T = 0.81 \text{N/m}$, $g = 10 \text{m/s}^2$]



5. A particle having mass m , charge q is projected with velocity v_0 along y -axis in a region of uniform magnetic field B_0 which is outward and perpendicular to the plane of the paper as shown in the figure. The particle is continuously subjected to a frictional force which varies with velocity as $\vec{F}_r = -\alpha\vec{v}$, where α is a constant. Consequently, the particle moves on a spiral path till it comes to rest at point P. Find the x -co-ordinate (in metre) of point P. (Take $\alpha = 10^{-3}$ kg/sec, $q = 10^{-3}$ C, $B_0 = 1$ T, $v_0 = 1$ m/s, $m = 20$ gm)



6. An inductor of 0.1 H and a box are connected to AC supply of 25 volts and $\omega = 100$ rad/s. Over all power factor of circuit is 1 and the reading of AC ammeter (I_{rms}) is 5 A. Find the power factor of box.



SECTION-III : (Maximum Marks: 18)

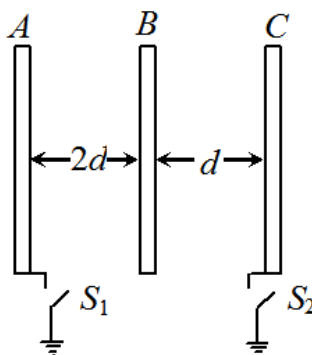
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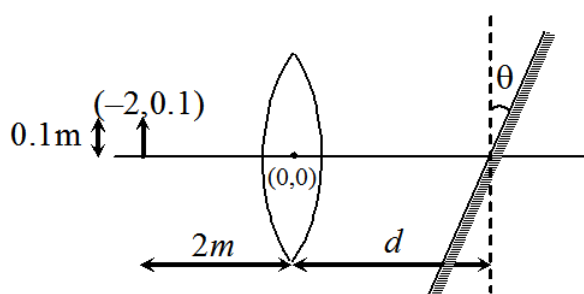
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

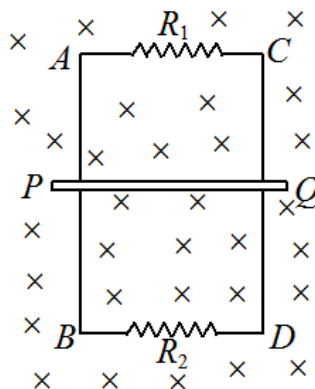
- Three identical, parallel conducting plates A, B and C are placed as shown. Switches S_1 and S_2 are open and can connect A and C to earth when closed. A charge $Q = -3 \mu\text{C}$ is given to B. If S_1 and S_2 both are closed together. Find the amount of charge in μC on inner surface of plate A.



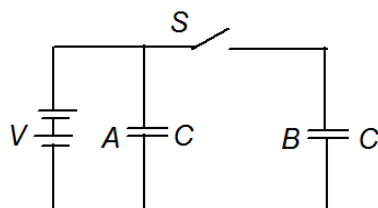
- A convex lens of focal length 1.5m is placed in a system of coordinate axis such that its optical centre is at origin and principal axis coinciding with the x-axis. An object and a plane mirror are arranged on the principal axis as shown in figure. Find the value of d (in m) so that y-coordinate of final image (after refraction and reflection) is 0.3m. (Take $\tan\theta = 0.3$)



3. Two parallel vertical metallic rails AB and CD are separated by 1m. They are connected at the two ends by resistances R_1 and R_2 as shown. A horizontal metallic bar PQ of mass 0.2 kg slides without friction, vertically down the rails under the action of gravity. There is uniform horizontal magnetic field of 0.6T perpendicular to plane of the rails. It is observed that when the terminal velocity attained, the power dissipated in R_1 and R_2 are 0.76 W and 1.2 W respectively. Find the terminal velocity of bar in m/s. ($g = 9.8 \text{ m/s}^2$)

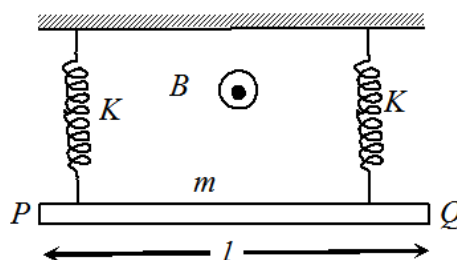


4. The figure shows two identical parallel plate capacitors connected to a battery with switch S closed. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant (or relative permittivity) 3. The ratio of the total electrostatic energy stored in both capacitors before and after the introduction of the dielectric is $\frac{x}{5}$. Then find the value of x



5. A particle is projected vertically upwards from the surface of the earth with a kinetic energy equal to $\frac{1}{3}$ times the minimum kinetic energy needed to escape. If radius of the earth is 6400km. The maximum height attained by the particle from the surface of the earth is $\frac{1}{N}$ times the radius of earth. Determine the value on N.
6. A conducting rod of mass m and length ℓ is connected by two identical springs as shown in the figure. Initially the system is in equilibrium. A uniform magnetic field of magnitude B directed perpendicular to the plane of the paper outwards also exists in the region. If a current I is switched on that passes from P to Q through the rod. Maximum elongation in the spring is $\frac{BI\ell}{NK}$. Determine the value of N.

[Given: $|mg| = |BI\ell|$]



PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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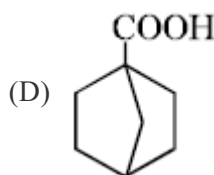
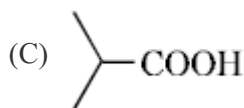
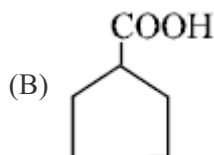
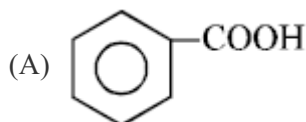
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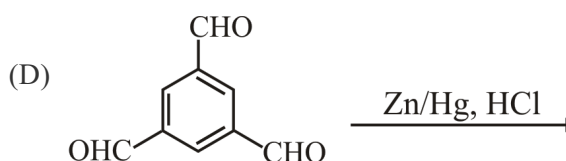
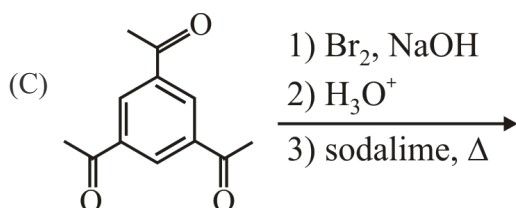
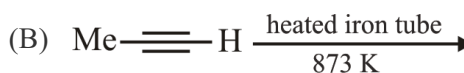
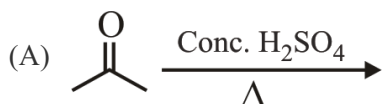
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

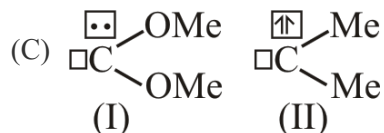
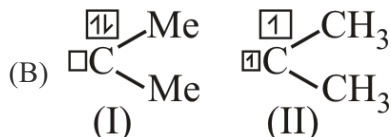
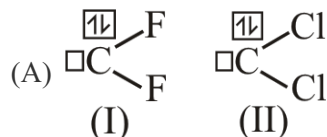
1. Which one of the following compounds will give HVZ reaction?



2. The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)



3. In which of the following Ist is more stable than IInd



(D) None

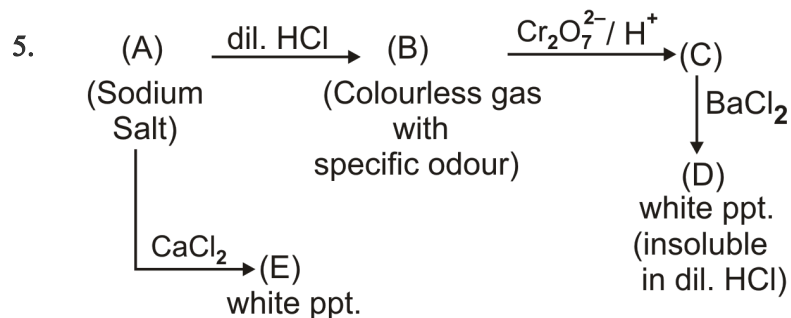
4. Find the incorrect order in the following.

(A) He < Ne < Ar < Kr < Xe (boiling point)

(B) He > Ne > Ar > Kr > Xe (ionization energy)

(C) He < Ne < Ar < Kr < Xe (Solubility in water)

(D) He > Ne > Ar > Kr > Xe (melting point)



Identify the correct statement :-

(A) Salt A having $\text{S}_2\text{O}_3^{2-}$

(B) White ppt. (D) is BaSO_4

(C) B is H_2S

(D) C form yellow ppt. with HgCl_2

6. Which reactions produce Chlorine?

(A) $\text{KMnO}_4 (\text{aq.}) + \text{HCl} (\text{conc.}) \rightarrow$

(B) $\text{NaCl}_{(\text{aq.})} (\text{Brine}) \xrightarrow[\text{using Pt electrode}]{\text{Electrolysis}}$

(C) $\text{NaOCl}_{(\text{aq.})} + \text{HCl}_{(\text{aq.})} \rightarrow$

(D) $\text{NaCl}_{(\text{s})} + \text{H}_3\text{PO}_4 (\text{conc.}) \xrightarrow{\Delta}$

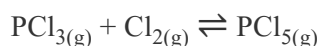
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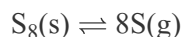
Zero Marks : 0 In all other cases.

1. At what temperature (in K) is the following reaction at equilibrium, when all substances are at standard pressure? Assuming that entropies and enthalpies of reaction do not vary with temperature.



	ΔH_f° (kJ/mole)	S° (J/mole-K)
$\text{PCl}_{3(g)}$	-290	310
$\text{Cl}_{2(g)}$	0	220
$\text{PCl}_{5(g)}$	-375	360

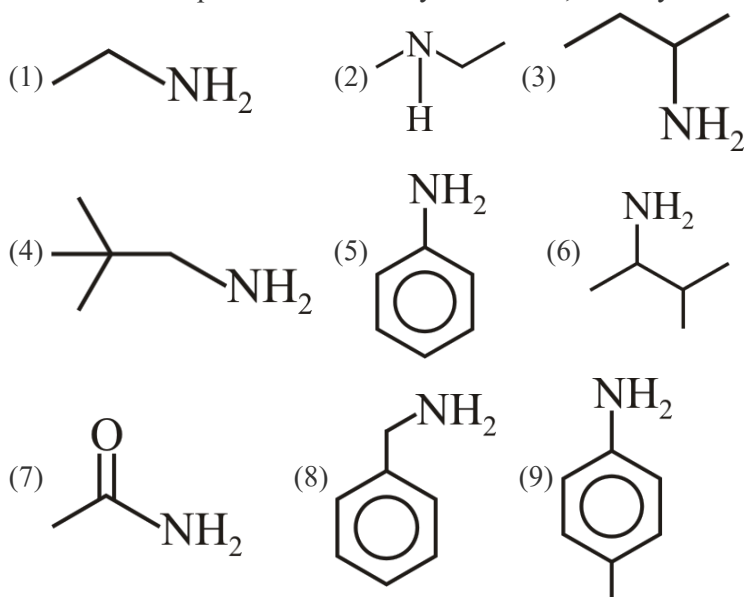
2. 2.56gm of sulphur $\text{S}_8(\text{s})$ is taken which achieves equilibrium with its vapours according to reaction



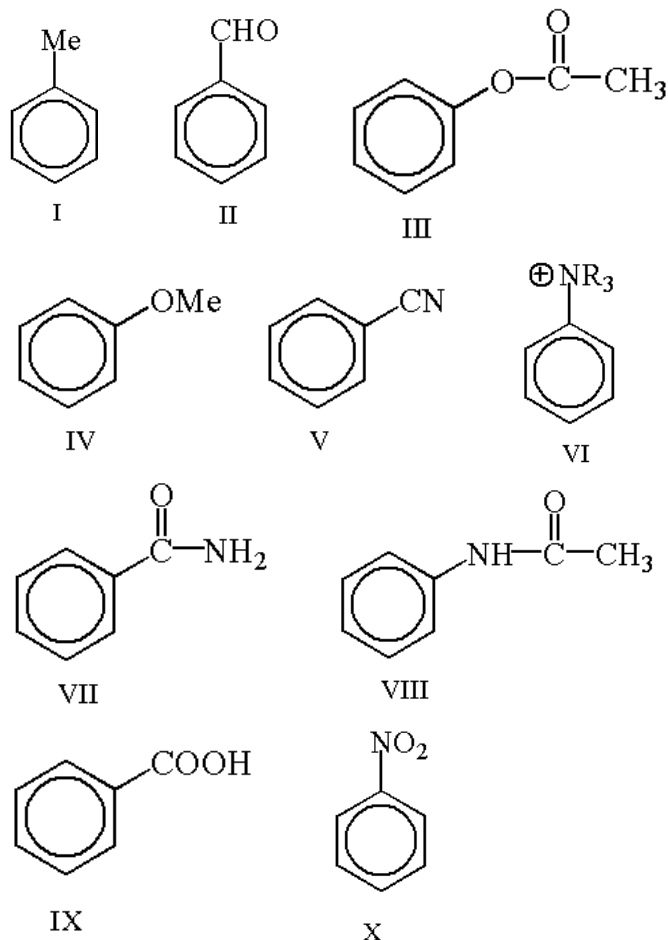
If vapours occupies 960 ml at 1 atm, 300K, then degree of dissociation of $\text{S}_8(\text{s})$ will be -

[$R = 0.08 \text{ atm L / mol-K}$] [Atomic mass : S = 32]

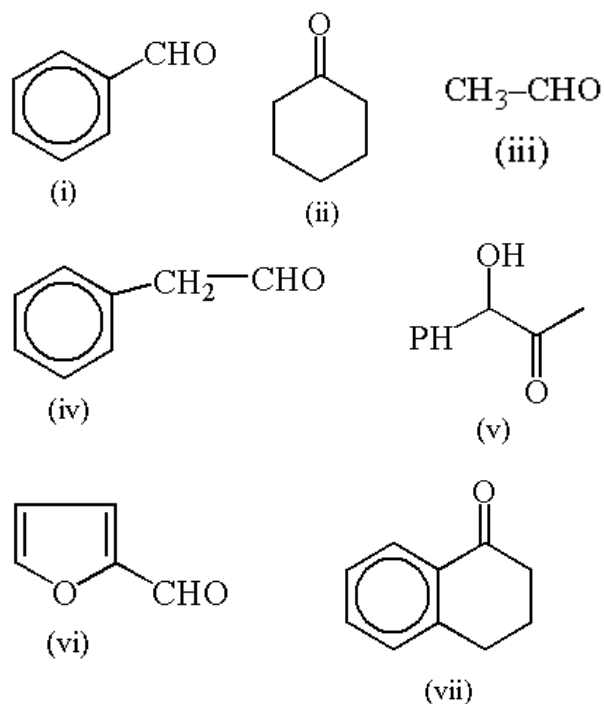
3. Number of compounds show Carbylamine test, or Isolynise test



4. Number of compound show more rate of EAS than benzene.



5. How many compound show Fehling and 2, 4 DNP test.



6. The rate constant of the reaction $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ is $3 \times 10^{-3} \text{ min}^{-1}$. At what concentration of H_2O_2 , (in M) the rate of the reaction will be $2 \times 10^{-4} \text{ M s}^{-1}$.

SECTION-III : (Maximum Marks: 18)

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- In a cyclic process ABCA consist of three processes. if $q_{AB} = -20\text{J}$; $q_{BC} = -40\text{ J}$; $q_{CA} = 30\text{ J}$ & $W_{AB} = 80\text{ J}$; $W_{BC} = -40\text{ J}$ then find work done by system in process CA.
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.
- Consider two reactions at 27°C
(I). $A \rightarrow B$; (I order kinetics)
(II). $X \rightarrow Y$; (Zero order kinetics)
If both A & X each having initial concentration 0.1M & same initial half life period. If ratio of rate constants for I & II reactions. $\left(\frac{K_I}{K_{II}}\right)$ is 'X' find value of $\left(\frac{X}{2}\right)$? [$\ln 2 = 0.7$]
- KF crystallizes in the NaCl type structure. If the radius of K^+ ion is $\frac{186}{\sqrt{2}}$ pm and F^- ion is $\frac{214}{\sqrt{2}}$ pm, what is the shortest distance between $K^+ - K^+$ ions (in Å in KF crystal) ?
- Find the maximum number of identical S-O bond length in the anionic part of $\text{Na}_2\text{S}_2\text{O}_8$.
- Find the number of moles of PH_3 formed when one mole of white phosphorous is treated with NaOH.
- Find the moles of Na_2SO_3 required for the reduction of one mole of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic solution.

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following function(s) is(are) surjective ?

[Note: $[m]$ and $\{m\}$ denotes greatest integer function and fractional part function respectively, and D_f denotes the domain of the function $y = f(x)$]

(A) $f : D_f \rightarrow \mathbb{R}, f(x) = \ln(\tan(\pi[x]) + |x^2 + 2x - 3|)$

(B) $g : D_g \rightarrow \mathbb{R}, g(x) = \frac{x^2 + 2x - 3}{x - 1}$

(C) $h : D_h \rightarrow \mathbb{R}, h(x) = \ln\left(\frac{1-x}{1+x}\right)$

(D) $k : D_k \rightarrow \mathbb{R}^+, k(x) = \sqrt{[x] + [-x] + 1} + \sqrt{\{x\} + \{-x\} + 1}$

2. There are n -students, namely $S_1, S_2, S_3, \dots, S_n$ and n -chairs placed in a row. Let Q_n denotes the number of ways of seating these n -students such that no two students having consecutive numbers sit together. Then

(A) $Q_3 = 0$

(B) $Q_4 = 2$

(C) $Q_5 = 14$

(D) $Q_{21} > 19Q_{20}$

3. Let f be a continuous function on $\mathbb{R}$ and satisfies $f(x) = e^x + \int_0^1 e^x f(t) dt$, then which of the following is(are) correct?
- (A) $f(0) < 0$ (B) $f(x)$ is decreasing function on $\mathbb{R}$
- (C) $f(x)$ is an increasing function on $\mathbb{R}$ (D) $\int_0^1 f(x) dx > 0$
4. If $\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx = \alpha \cdot x^\beta + \gamma \tan^{-1}(x^\delta) + C$, where C is constant of integration, then which of the following is(are) correct? $\{2\alpha, 3\beta, 4\gamma, 6\delta \in \mathbb{N}\}$
- (A) $\alpha = \frac{2}{3}$
- (B) $\alpha\beta = 1$
- (C) $(\alpha\beta + \gamma) = 7$
- (D) Sum of all coefficients of even powers of x in the expansion of $\left(\alpha\beta + \frac{x}{\delta}\right)^\gamma$ is equal to $\frac{(\gamma+1)^\gamma + (\gamma-1)^\gamma}{2}$.
5. The value of the definite integral $\int_{-1}^0 \left(\sqrt[3]{x^4 - 2x^2} - \sqrt{1 - \sqrt{x^3 + 1}} \right) dx$ lies in the interval
- (A) $\left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$ (B) $(-\pi, \pi)$
- (C) $\left(\frac{\pi}{4}, \frac{\pi}{2} \right)$ (D) $\left(-\frac{\pi}{3}, \frac{\pi}{3} \right)$
6. If $F(x) = \int e^x \left(\int_0^x x \sin x^2 dx + \int_0^x (\sin x^2 + 2x^2 \cos x^2) dx \right) dx$, $F(0) = \frac{-1}{2}$ then which of the following statement(s) must be true.
- (A) $F\left(\sqrt{\frac{4\pi}{7}}\right) < 0$ (B) $F'(0) = \frac{-1}{2}$
- (C) $F''(0) + F'''(0) = -1$ (D) $F(0) - F'(0) = \frac{1}{2}$

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. $f(x)$ is continuous and bijective function on $\mathbb{R}$. If $\forall a \in \mathbb{R}$, the area bounded by $y = f(x)$, $x = 3 - a$, $x = 3$ and x -axis is equal to area bounded by $y = f(x)$, $x = 3 + a$, $x = 3$ and x -axis. Then the value of k if $\int_{-\lambda}^{\lambda} f^{-1}(x) dx = k\lambda$ is
2. If $f'(x) = \phi(x)$ and $\phi'(x) = f(x) \forall x \in \mathbb{R}$, also $f(3) = 5$ and $f'(3) = 4$, then value of $(f^2(10) - \phi^2(10))$ is
3. Consider a twice differentiable function $y = f(x)$ satisfying $f(x) + f''(x) = 2f'(x)$ where $f(0) = 0$, $f(1) = e$ then find $\int_0^1 f(x) dx$
4. Given $p, q \in \{0, 1, 2, 3, 4, \dots, 19, 20\}$. Consider the system of equations

$$\begin{aligned} x + y + z &= 4 \\ 2x + y + 3z &= 6 \\ x + 2y + pz &= q \end{aligned}$$
 Let L : denotes number of ordered pairs (p, q) so that the system of equations has a unique solution,
 M : denotes number of ordered pairs (p, q) so that the system of equations has no solution and
 N : denotes number of ordered pairs (p, q) so that the system of equations has infinite solutions.
 Find $(L + M - N)$.
5. Let f be a differentiable function satisfying $f'(x) = f^2(x)$ and $f(0) = -1$.
 If $\frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1} \right) dx = \frac{p}{q}$ where $p, q \in \mathbb{N}$ then find least value of $(p+q)$.
6. If area of the region whose upper left boundary is enclosed by the curve $f(x) = \sin x + x + 2$, lower left by the line $x + y = 2$, lower right by $f^{-1}(x)$ and upper right by the line $x + y = 2\pi + 2$ is $k(\pi + 1)$ then find the value of k

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

- Let $y = f(x)$ be the curve passing through $M(0, 2)$ such that the sum of coordinates of any point $P(x, y)$ on the curve exceeds the slope of tangent to the curve at that point by 5. Find the number of distinct real solutions of the equation $f'(x) = 0$.
- Let $\alpha(t)$ and $\beta(t)$ be differentiable functions on $\mathbb{R}$ such that $\alpha(0) = 2$ and $\beta(0) = 1$. If $\alpha(t) + \beta'(t) = 1$ and $\alpha'(t) + \beta(t) = 1$ for all $t \in [0, \infty)$, then the value of $\alpha(\ln 2)$ is expressed in the lowest form as $\frac{p}{q}$. Find the value of $(p - q)$.
- A square matrix of size $n \times n$ is called an orthogonal matrix iff $AA^T = A^T A = I_n$, $n \in \mathbb{N}$.
If $A = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ is an orthogonal matrix, then magnitude of $(a + b + c)$ is equal to
- Let A and B are orthogonal matrices of order 3×3 such that $X = AB^T$ and $Y = BA^T$, then the value of $\text{tr}(XY)$ is equal to -
- Let matrix $A = \begin{bmatrix} \alpha^2 & 6 & 8 \\ 3 & \beta^2 & 9 \\ 4 & 5 & \gamma^2 \end{bmatrix}$ and $B = \begin{bmatrix} 2\alpha & 3 & 5 \\ 2 & 2\beta & 6 \\ 1 & 4 & 2\gamma - 3 \end{bmatrix}$. If $\text{tr}(A) = \text{tr}(B)$ then value of $\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)$ is equal to (where $\text{tr}(A)$ denotes trace of A and α, β, γ are non-zero real numbers)
- W_1, W_2 and W_3 are three ticket windows in a cinema hall. 12 persons $(A_1, A_2, \dots, A_{12})$ are standing for the tickets in a queue of 4, 3 and 5 number of persons in front of W_1, W_2 and W_3 respectively. A_1 and A_2 want to get their tickets from W_1 and A_3, A_4 and A_5 want to get their tickets from W_3 . If the number of ways in which all are getting their tickets such that A_1, A_2, A_3, A_4 and A_5 are getting their tickets as early as possible is $k(7!)$, then find the value of $\left\lceil \frac{k}{3} \right\rceil$.
(Assuming each person is getting exactly one ticket at a time).
[Note: $[y]$ denotes greatest integer less than or equal to y .]

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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